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Name ________________________ Nov. 11, 2005 Sunday

Department ___________________

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University Physics

Midterm Examination

School of Intensive Instruction in Sciences and Arts, Nanjing University

Physical Constants

Electron charge	$e = 1.60\times 10^{-19}$ C
Dielectric constant of free space	$\varepsilon_0 = 8.85\times 10^{-12}$ F/m
Permeability of free space	$\mu_0 = 4\pi\times 10^{-7}$ N/A^2
Vacuum speed of light	$c = 3.00\times 10^{8}$ m/s
Electron mass	$m_e = 9.11\times 10^{-31}$ kg
Proton mass	$m_p = 1.67\times 10^{-27}$ kg
Electron volt (eV)	1eV=1.60×10^{-19} J

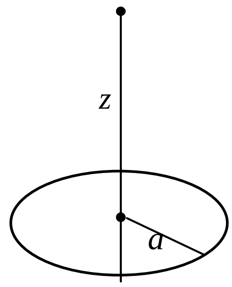
Select five out of following six problems.

1. The electric charge density of a spherically symmetric charge distribution is

$$\rho(r) = \begin{cases} \rho_0 (1 - \dfrac{r^2}{a^2}), & r < a, \\ 0, & r > a, \end{cases} \qquad (a > 0).$$

(a) Calculate the total charge Q.
(b) Find the electric field $\boldsymbol{E}$ and the electric potential $V(r)$ in the region $r > a$.
(c) Find the electric field $\boldsymbol{E}$ and the electric potential $V(r)$ in the region $r < a$.

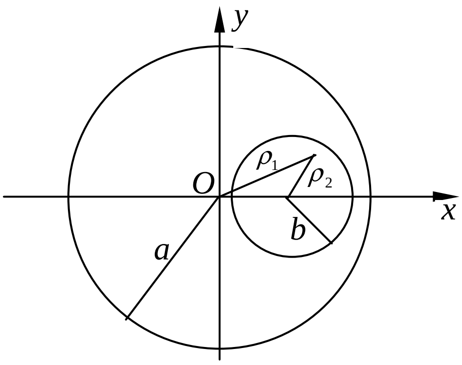
2. A circular disk of radius a has the surface charge density σ.



(a) Compute the electric potential at a point on the axis of symmetry and at distance z from the center of the disk and then the electric field there.

(b) Find the discontinuity of the electric field at the center of the disk. Reobtain the result by using the Gauss's law of the electric field.

3. A steady electric current I in the z-direction flows uniformly in the region between two cylindrical surfaces $x^2+y^2=a^2$ and $(x-a/2)^2+y^2=b^2$, where $a>0$, $b>0$ and $b<a/2$.



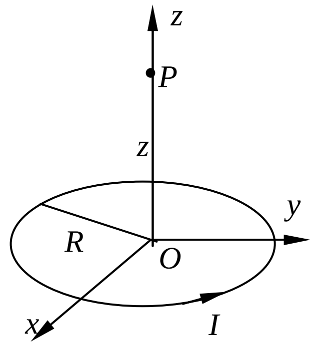
(a) Find the magnitude and direction of the magnetic field $\boldsymbol{B}$ at every point P on the x-axis.

(b) Show that the magnetic field $\boldsymbol{B}$ is uniform in the region $(x-a/2)^2+y^2<b^2$ and the field is

$$\boldsymbol{B}=\frac{\mu_0 I}{4\pi}\frac{a}{a^2-b^2}\boldsymbol{e}_y .$$

(Hint: A vector in the x-y plane that is perpendicular to the displacement $\boldsymbol{\rho}$ in the x-y plane and of the magnitude $\rho=|\boldsymbol{\rho}|$ can be expressed as $\pm\boldsymbol{e}_z\times\boldsymbol{\rho}$.)

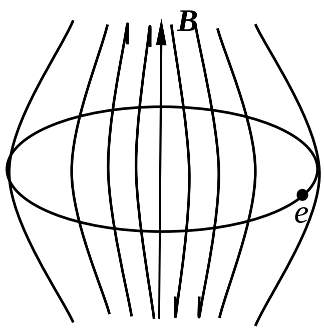
4. As shown in the accompanying figure, a loop of radius R carries an electric current I.



(a) Find the magnetic field at point P that is on the symmetric axis of the loop and at distance z from the center of the loop.

(b) If a magnetic dipole with magnetic moment $\boldsymbol{m} = m_0 \boldsymbol{e}_z$ is placed at point P, what is the magnitude and direction of the force acted on the dipole?

5. The betatron is a cyclic induction electron accelerator. In a betatron, the electron is accelerated by the circular electric field induced by the changing magnetic field that is axially symmetric. Simultaneously, the magnetic field also provides the centripetal force for the electron to move in a circular orbit. The poles of the magnet are tapered to cause the axially symmetric magnetic field to weaken with increasing radius. Show that if the average magnetic field inside the orbit



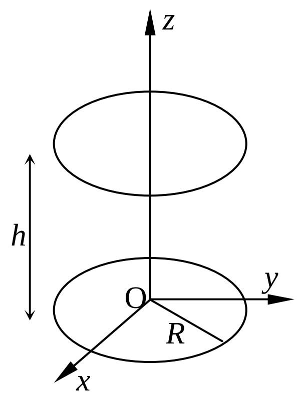
$$\overline{B} = \frac{1}{\pi R^2} \int_0^R B(\rho) 2\pi\rho d\rho$$

is always twice as strong as the magnetic field on the orbit $B(R)$, i.e., $\overline{B} = 2B(R)$, the electron can be accelerated stably on the orbit with fixed radius R.

6. A non-uniform time-varying electric field $\boldsymbol{E}$ given by

$$\boldsymbol{E} = E_0\left(1-\frac{\rho}{R}\right)\sin(\omega t)\,\boldsymbol{e}_z$$

is distributed in the cylindrical region of radius R and height h as shown in the figure.



(a) Find the displacement current density j_d in the cylindrical region ($\rho < R$).

(b) Find the magnetic field $\boldsymbol{B}$ induced in the same region.

(c) Indicate on the figure the direction of $\boldsymbol{B}$ at radial distance ρ from the axis at $t = 0$.